



## RS004 Week 16 — English Worksheet

**Topic:** Title · Level Select · Sound · Showcase (Final) · **Course:** Platformer Game · **Time:** about 45 minutes

This is the final week. You built a finished game over 16 weeks: a **title screen**, a **level select** menu, and **sound** for jumps and coins, all running on a **game state machine** that moves between title → level select → playing → game over / victory. This worksheet is about reading and explaining that code in plain English, then presenting your finished game.

Keep these words handy: **state machine**, **title screen**, **level select**, **sound**, **showcase**.

### 1 · Predict 🧠

Read each snippet. Write what you think it does.

```
current_state = "title"
# "title" → "level_select" → "playing" → "gameover" / "victory" → "title"
```

The game starts in **"title"**. List one path it can take through the states to a win.

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```
if current_state == "title":
    if event.key == pygame.K_SPACE:
        current_state = "playing"
    elif event.key == pygame.K_l:
        current_state = "level_select"
```

On the title screen the player presses L. What screen comes next?

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```
if event.key == pygame.K_SPACE and on_ground:  
    velocity_y = JUMP_STRENGTH  
    jump_sound.play()
```

**When does the jump sound play? Could it play in the air? Why or why not?**

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## 2 · Map the States

Write out the arrows showing which state leads to which. Fill in the blanks.

```
title  --space--> _____  
title  --L-----> _____  
level_select  --Enter--> _____  
playing  --hit enemy--> _____  
playing  --clear last level--> _____  
gameover  --R--> _____
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### 3 · Spot the Bug 🐛

Each snippet is broken. On paper, rewrite it so it works, then explain why the original was wrong.

**Bug A** — Pressing space on the title should start the game. Right now space does nothing because the key is only checked while playing.

```
if current_state == "playing":
    if event.key == pygame.K_SPACE:
        velocity_y = JUMP_STRENGTH
```

**Hint:** the title screen also needs to react to space. Handle each state's keys.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — In the level select menu, ↑/↓ should move the highlight and wrap around the ends. Right now going past the last item crashes with an index error.

```
if event.key == pygame.K_DOWN:
    selected_level = selected_level + 1
```

**Hint:** use `% len(LEVELS)` so the choice wraps back to the start.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — A coin pickup should play a sound, but the game is silent even though `coin_sound` exists.

```
for coin in coins[:]:
    if player.colliderect(coin):
        coins.remove(coin)
        score += 1
```

**Hint:** the sound is created but never played on pickup.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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## 4 · Explain the Code

This is the heart of your finished game — the state machine that decides what screen the player sees and what each key does.

```
current_state = "title"

while running:
    for event in pygame.event.get():
        if event.type == pygame.KEYDOWN:
            if current_state == "title":
                if event.key == pygame.K_SPACE:
                    current_state = "playing"
                elif event.key == pygame.K_l:
                    current_state = "level_select"
            elif current_state == "level_select":
                if event.key == pygame.K_RETURN:
                    current_state = "playing"
            elif current_state == "gameover":
                if event.key == pygame.K_r:
                    current_state = "title"

    if current_state == "title":
        draw_title()
    elif current_state == "level_select":
        draw_level_select()
    elif current_state == "playing":
        update_player()
        if player_died:
            current_state = "gameover"
    elif current_state == "gameover":
        draw_gameover()
```

What value does `current_state` start as, and what screen does the player see first?

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On the title screen, what do the SPACE key and the L key each do?

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Inside the "playing" state, what makes the game switch to "gameover" ?

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From "gameover", which key sends the player back, and back to which state?

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Why does the code check `current_state` both in the event loop and again lower down when drawing?

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## 5 · Explain Your Lesson Code 🎥

This is your final presentation. Record a phone video of the **finished platformer YOU built across the course**. Show it running from title to victory, then explain the key parts of your own code. Try to use these words: **state machine, title, level select, sound, function**.

1. Show the title screen, then level select, then play through a level with sound.
2. Open your code and point to your state machine — explain how it moves between screens.
3. Show one function or class you are proud of and explain what it does.
4. Reach the victory screen, and say the hardest bug you fixed and how.

**Write what you will say in your video. Plan it here before you record.**

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**Submit** 

Send this worksheet + your finished-game presentation video to teacher on KakaoTalk.